

AUTOMATED DEFECT DETECTION AND REJECTION SYSTEM

Research Report

Robotics & Automation Internship — Task 3

CodeAlpha

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1. INTRODUCTION

In modern manufacturing, quality assurance is critical to maintaining brand reputation and operational efficiency. This report details the design of an automated conveyor belt system that utilizes a Programmable Logic Controller (PLC) and integrated sensors to identify and sort products based on physical and aesthetic specifications. The system is designed to operate autonomously, reducing the need for manual inspection while increasing throughput.

2. SYSTEM CONCEPT AND IDEA

The core idea behind this design is a Closed-Loop Quality Control (QC) mechanism. Instead of a simple transport belt, this system acts as a decision-maker. It evaluates every item against two specific criteria: Dimensional Accuracy and Color Consistency. By using a PLC as the central processing unit, the system can make split-second decisions to either allow a product to proceed to the packaging stage or physically intervene to remove a defective unit.

3. PROCESS DESCRIPTION (FLOWCHART ANALYSIS)

The operational logic follows a structured sequence as outlined in the system flowchart:

- **Startup and Diagnostics:** The process begins with a Power-On Self-Test (POST). The PLC checks all connected I/O modules, sensors, and actuators. If a hardware fault is detected, the system triggers a Fault Alarm and halts to prevent mechanical damage.
- **Detection Phase:** Once initialized, the conveyor motor starts. An Infrared (IR) sensor scans the belt on a 10ms cycle. When an object breaks the sensor beam, the inspection sequence is triggered.
- **Dual-Stage Inspection:**
 - **Geometry Check:** The system first measures the object's dimensions. If the size falls outside the Min/Max thresholds (Geometric Defect), a flag is raised.
 - **Color Analysis:** Simultaneously, an RGB color sensor scans the surface. The PLC processes these values to ensure they match the reference product color.

- **Sorting Logic:**
 - **Pass:** If both size and color are within range, the item continues to the "Accept Bin," and the software increments the Accept Counter.
 - **Fail:** If either test fails, the PLC activates a mechanical arm. This arm sweeps the defective item into a "Reject Bin," and the Reject Counter is updated.
- **Safety and Data Logging:** The system continuously monitors for an Emergency Stop (E-Stop). At the conclusion of a batch, the PLC generates a Shift Report, logging all timestamps and defect data to memory before a controlled shutdown.

4. TECHNICAL FUNCTIONALITY

The system bridges the gap between digital logic and physical actuation. The PLC interprets low-voltage signals from the IR and RGB sensors and uses internal logic gates to determine the product's status. To handle the physical rejection, the PLC sends a signal to a motor driver or pneumatic controller that powers the mechanical arm. This separation of logic and power ensures that the sensitive controller is protected from the high-current demands of the motors.

5. CONCLUSION

The Automated Defect Detection and Rejection System offer a robust solution for high-speed production environments. By integrating real-time sensing with automated mechanical sorting, the design ensures that only products meeting strict quality standards reach the consumer. The addition of data logging further allows for process optimization by identifying recurring patterns in production defects.